

ARAK, Iran, Islamic Republic of

WMO: 407690

Lat: 34.0716N

Lon: 49.7837E

Elev: 1703

StdP: 82.47

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-9.4	-17.6	1.0	-6.2	-14.7	1.3	0.7	10.1	6.8	8.8	8.2	0.3	120	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	36.7	15.9	35.5	15.5	34.4	15.2	18.2	32.1	17.1	31.8	16.3	31.4	3.9	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.2	11.7	25.6	11.4	10.3	23.5	10.0	9.4	22.1	58.9	31.6	54.9	31.8	52.1	31.2	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
(a)	(b)	(c)		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
9.2	7.9	6.8	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
			WB	-14.6	38.7	5.1	1.2	-18.3	39.6	-21.3	40.3	-24.1	40.9	-27.8	41.8	
				-15.4	19.5	5.0	2.1	-19.0	21.1	-21.9	22.3	-24.7	23.5	-28.3	25.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	0.2	3.3	8.4	13.3	18.8	24.8	27.8	26.8	22.0	15.7	7.8	3.1		
	DBStd	10.01	5.79	4.91	4.13	3.40	2.85	2.51	2.05	1.75	2.29	3.15	3.69	3.83		
	HDD10.0	882	303	189	79	11	0	0	0	0	0	1	85	214		
	HDD18.3	2353	561	420	308	153	29	1	0	0	2	92	317	471		
	CDD10.0	2488	0	2	29	110	274	443	552	519	359	178	18	1		
	CDD18.3	918	0	0	1	2	44	194	294	261	112	10	0	0		
	CDH23.3	11673	0	0	2	26	486	2360	3913	3356	1367	163	0	0		
	CDH26.7	5809	0	0	0	1	106	1136	2204	1816	531	15	0	0		
(14) Wind	WSAvg	2.0	1.4	2.0	2.6	2.6	2.6	2.4	2.4	2.0	1.9	1.8	1.3	1.2		
(15) Precipitation	PrecAvg	266	31	31	48	44	21	3	2	2	1	15	32	36		
	PrecMax	393	65	82	103	105	78	20	9	25	19	56	127	121		
	PrecMin	160	0	3	0	4	0	0	0	0	0	0	0	0		
	PrecStd	62	19	19	27	30	20	5	2	5	4	16	28	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	16.6	23.2	26.2	31.0	36.5	38.5	37.3	33.7	28.2	20.9	16.6		
		MCWB	4.9	6.8	9.7	11.5	13.6	15.3	16.3	16.5	14.3	11.9	9.0	6.0		
	2%	DB	11.5	14.1	20.4	23.6	29.2	34.9	37.3	36.0	32.5	26.7	18.6	13.6		
		MCWB	4.2	5.7	8.1	10.8	12.9	14.8	15.9	15.7	14.0	11.5	7.8	5.6		
	5%	DB	9.6	12.3	18.1	22.0	27.7	33.6	36.1	35.0	31.3	25.2	16.7	11.3		
		MCWB	3.6	4.8	7.3	10.1	12.4	14.5	15.8	15.2	13.4	10.9	7.6	4.8		
	10%	DB	7.8	10.5	16.0	20.3	26.2	32.2	34.9	34.0	29.9	23.5	14.6	9.3		
		MCWB	2.7	4.1	6.5	9.6	12.0	14.1	15.7	15.0	12.9	10.3	7.0	4.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	8.0	10.9	13.1	15.6	18.4	20.5	20.2	16.8	13.7	10.7	7.7		
		MCDB	10.8	13.7	20.5	21.7	26.4	31.1	32.1	34.2	30.7	24.7	16.8	12.8		
	2%	WB	5.2	6.6	9.1	12.1	14.6	16.9	18.7	17.7	15.2	12.5	9.5	6.4		
		MCDB	9.7	12.6	17.3	21.0	25.8	30.4	32.0	32.9	29.9	23.3	15.7	11.4		
	5%	WB	4.1	5.5	8.0	11.2	13.8	15.8	17.5	16.7	14.2	11.7	8.5	5.4		
		MCDB	8.6	10.9	16.1	19.7	24.6	30.2	32.1	31.8	28.8	22.3	14.4	10.1		
	10%	WB	3.0	4.5	7.1	10.4	13.0	15.0	16.6	15.9	13.5	10.9	7.6	4.4		
		MCDB	7.4	9.7	14.4	18.3	23.3	29.8	31.9	31.5	27.9	21.1	13.1	8.6		
(35) Mean Daily Temperature Range	MDBR		9.3	10.2	11.3	11.3	12.6	14.0	13.9	14.8	15.9	14.1	11.0	9.6		
	5% DB	MCDBR	11.9	12.7	13.7	13.6	14.5	15.5	15.2	16.1	17.2	16.0	14.0	12.7		
		MCWBR	6.7	6.5	6.0	5.0	5.0	5.5	5.2	5.7	6.5	6.7	6.5	7.1		
	5% WB	MCDBR	10.8	11.4	11.9	12.0	13.1	14.0	13.5	14.8	16.0	14.6	11.7	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.287	0.324	0.380	0.469	0.463	0.449	0.489	0.438	0.374	0.381	0.326	0.287		
	taud		2.414	2.304	2.123	1.907	1.946	1.993	1.901	2.033	2.206	2.157	2.337	2.453		
	Ebn at Noon		928	920	888	823	831	840	803	840	884	840	867	907		
	Edh at Noon		90	113	148	193	188	179	195	168	134	130	97	83		
(44) All-Sky Solar Radiation	RadAvg		2.94	3.85	5.00	5.81	6.94	8.19	7.75	7.28	6.30	4.63	3.22	2.65		
	RadStd		0.24	0.32	0.40	0.48	0.49	0.32	0.29	0.25	0.23	0.29	0.29	0.32		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.72	-0.85	-1.21	N/A	N/A	N/A	N/A

Nomenclature: See separate page